Abhishek Singh

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Research Interests

Machine Learning, Computer Vision, Deep Learning, Robotics

Academic History

Year	Qualification	Institute	Performance	Subjects
2015-Present	B.Tech, Electrical Engineering	Indian Institute of Technology (B.H.U, Varanasi	9.09/10	Project-Localization of Mobile Robots
2015	XII (C.B.S.E)	D.A.V Public School, Kota	92.2/100	Mahematics-97/100
2013	X (I.C.S.E)	St. Peter's School, Panchgani	91.5/100	Computer Science- 98/100

Skills and Certifications

- **Programming/ Scripting Languages**- C, C++, Python, Java, CSS, PHP, HTML
- **Software/Libraries**- Tensorflow, Keras,scikit-learn, CUDA, Git, Grafana, Prometheus, Data Structure and Algorithms
- **Machine Learning**-SVM,Decision Trees, Ensemble Learning, Dimensionality Reduction, CNN, RNN, Deep Neural Networks
- **MOOCs**- Deep Learning Specialization, Sequence Models, CNNs, Hyper-Parameter Tuning, Optimization, Structuring Machine Learning Projects, Neural Network, and Deep Learning, Machine Learning-Coursera

Research, Teaching and Internship Experience

Research Internship

Design and Innovation Hub at IIT(BHU) Varanasi **Prof.R.K.Misra** **May-July 2017**

- Developed Solar Energy Forecasting model using Tensorflow, applied LSTM and Ridge Regression, with accuracy of 16% MAPE.
- Applied Random Forest, LightGBM and Ridge Regression on the dataset from Kaggle Solar Forecast Challenge to compare accuracy.

Design and Innovation Hub at IIT(BHU) Varanasi **Prof.Devender Singh** **May-July 2016**

- Designed a Smart Metering System, using GSM and Arduino UNO, and created a website for Electricity Users for Usage analysis.
- Features of this model are Online Monitoring through webpage, Electricity Cut-off and Electric Billing through an SMS using GSM.

Teaching Assistant

1. Basic Electrical Engineering- Teaching and Paper Correction (Under Graduate Sophomore Level Course) **July- December 2018**
2. Computer Programming and Linux Concepts-Teaching and Paper Correction (UG freshman Level Course) **January-May 2018**

Software Engineering Internship at Indus Valley Partners, Noida, India

- Created a Resource Monitoring System (Cloud watch) Using Prometheus, Grafana and Alert manager to Send Alerts, Monitor Servers on Grafana Dash Boards.Link to Project- https://github.com/asking28/resource_monitoring/wiki
- Features developed include Sending Email alerts on Server Over-Usage, Monitoring through Prometheus and Grafana dashboards.

Honors, Awards, and Presentations

1. Poster and Project Presentation at 100th Institute Day IIT (BHU). **2017**
2. Project Presentation at IIT Guwahati technical festival Techniche. **2016**
3. Awarded KVPY fellowship, Research Scholarship by Indian Government with top 99.4 percentile **2015**
4. Cleared IIT-JEE Advanced and IIT-Mains with top 99.99 Percentile. **2015**

Projects

Mobile Robot Localization

- Applied Kalman Filtering on for positioning using Accelerometer and four Ultrasound sensors.

Object Recognition and Classification

- Object detection using YOLOv2 and YOLOv3 and YOLOv3_tiny architectures using Python and Tensorflow. Trained on Custom dataset and increased data using Tensorflow data Augmentation. [Project]
- Evaluated Recall metrics of above architectures, with YOLOv3 being highest at 70% on testing images.

Electricity Price Forecasting

- Price forecasting with Neural Networks on MATLAB, MAPE of 6.5%, and MAE of 5.2%.
- Analyzed hourly and weekly demands of Electricity Load and their corresponding prediction errors.

Relevant Coursework

Probability and Statistics	Mathematical Modeling	Data Mining
Calculus	Ubiquitous Computing	Mathematical Methods
Numerical Techniques	Machine Learning	

Extracurricular Activities

1. Won Gold Medal in Javelin Event in Spardha Sports Meet 2017, biggest sports festival in North India.
2. Served as Vice- Captain of Athletics Team at IIT(BHU) and lead the team at Inter-IIT Meet at IIT Madras.
3. Participated in Inter-IIT tournament at IIT-Kanpur and IIT- Madras for Javelin Event.
4. Participated in Blood Donation camps organized in college. Mentored Students with Under Privileged background.
5. Core team member of Drone Tech Event at Technex annual Techno-Management festival and Aero Modeling club of IIT (BHU).